Solutions and teaching notes, chapter 2

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $x^2 - 6x + 9$. *Action:* Reconstruct the product of two binomials.
2. $(x - 4)(x + 4)$. *Action:* Use the difference of two squares on two examples.
3. x = 2 or $x = -5$. *Action:* Have students verbalise the zero-product rule and its assumption.
4. $x = -3$ or $x = 3$. *Action:* Distinguish the positive square root from the solutions of the equation.
5. Negative before 3, zero at 3, positive after 3. *Action:* Draw the line then make the sign table.
6. 4 attained only at 1. *Action:* Start from the fact that a square is non-negative.

Investigation

The positive lengths x and y satisfy $x + y$ = 10 and xy = 24. They are the roots of $t^2 - 10t + 24 = (t - 4)(t - 6)$. Dimensions 4 m and 6 m; check the area and the perimeter.

Direct applications

1. $g(x) = (x + 1)^2 - 4 = (x - 1)(x + 3)$. Roots 1 and -3, minimum -4 at -1.
2. The first is 3($x - 1)^2$ = 0, so 1. For the second, Delta= $-16 < 0$: no real solution.
3. $f(x) = a(x - 1)(x - 4)$; $4a = -8$, so $f(x) = -2(x - 1)(x - 4)$.
4. $(x - 5)(x + 1) < 0$ gives] -1; 5[. The vertex of $x(8 - x)$ is at 4, outside the domain; on [0; 3], the function increases and its maximum is 15 at 3.

Exercises: methods and answers

Exercise 2.1 - Completing a square. $x^2 - 8x + 7 = (x - 4)^2 - 9 = (x - 1)(x - 7)$. The leading coefficient is positive; the quadratic is negative or zero between the roots: [1; 7].

Exercise 2.2 - Discussing without heavy calculation. $x^2 + 4 > 0$ always, no root. ($x + 2)^2 \geq 0$, double root -2. For the last one, roots -4 and 1; positive on] $-$ 4; 1[, negative outside, zero at the bounds.

Exercise 2.3 - Choose a strategy. Identity: $x = \pm 3$. Sum and product: 4 and 5. For the third, no obvious integer root: Delta= 17 and $x = \frac{3 \pm \sqrt{17}}{4}$. The discriminant applies in all cases, but is not always the most efficient.

Exercise 2.4 - Finding a parabola. $f(x) = a(x + 2)(x - 3)$. At 1: $-6a$ = 12, therefore $a = -2$; $f(x) = -2x^2 + 2x$ + 12, hence $f(0) = 12$.

Exercise 2.5 - A vegetable garden against a wall. The two widths use 2 x metres; length $20 - 2x$. Domain [0; 10] for the function,]0; 10[for a non-degenerate rectangle. $A(x) = 20x - 2x^2 = 50 - 2(x-5)^2$. Maximum 50 m ^{2}, dimensions 5 m and 10 m.

Exercise 2.6 - A parameter. $\Delta = 4 - 4m$. Two roots if $m < 1$; double root 1 if $m = 1$; none if $m > 1$. The parameter m stays fixed when solving for x.

Exercise 2.7 - Link with sequences. $u_n = (n-1)(n-4)$; negative for $1 < n < 4$. Since n is an integer, the only indices are 2 and 3. The set of indices is not the whole real interval.

Exercise 2.8 - Delayed review. The minimum is -5 at -3. The inequality gives $(x+3)^2 \leq 1$, i.e. $-1 \leq x+3 \leq 1$, so $[-4; -2]$.

CQFD, exercise 43 p. 99, questions a and c. a: obvious root 1, product -30, hence the second root -30. c: write 4 $x^2 + 9x + 5$; root -1, product $\frac{5}{4}$, hence the second root $-\frac{5}{4}$. Check the sums, respectively -29 and $-\frac{9}{4}$.

CQFD, exercise 49 p. 99, questions a and b. The numbers are the roots of $t^2 - st + p$. a : $t^2 - t + 1$, discriminant -3: no real pair. b: $t^2 + 60t + 756$, discriminant 576 = 24 ^{2} : -18 and -42. Their sum is -60 and their product 756.

MCQ: answers and reasons

MCQ 1.

1. C. Set each factor to zero.

2. A. $4 - 12 = -8$.

3. D. The square vanishes at 2, value 5.

4. B. $-\frac{b}{a} = \frac{6}{2} = 3$.

5. C. Positive leading coefficient, strict inequality.

MCQ 2.

1. D. Algebraic identity.

2. A. $\frac{c}{a} = -\frac{2}{3}$.

3. C. The function increases up to 2; on this domain, right endpoint 1.

4. B. For a nonzero, a zero discriminant characterises a double root.

5. D. The multiplier is arbitrary and nonzero.

True or false

1. False: $x^2 + 1$ is strictly positive.

2. False without an assumption on a; $-(x-1)(x-2)$ is positive between 1 and 2.

3. True: the square of a real number is positive or zero; equality only at 2.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.